signal.sci (C:\Users\LENOVO\Documents\signal.sci) - SciNotes

*signal.sci

```
// Evaluate X(z) on the unit circle
for i = 1:length(z_vals)
z_val = z_vals(i);
X_z_val = 0;
for j = 1:length(n)
X_z_val = X_z_val + x_n(j) * z_val^(-n(j)); // Evaluate X(z) at z_vals(i)
end
magnitude(i) = abs(X_z_val); // Magnitude
// Calculate phase using atan
phase(i) = atan(imag(X_z_val) / real(X_z_val));
if real(X_z_val) < 0 then
phase(i) = phase(i) + %pi;
end
end
// Plot magnitude and phase of the Z-transform
clf; // Clear the current figure
// Plot magnitude
subplot(2,1,1);
plot(theta, magnitude, 'b');
xlabel('Theta (radians)');
ylabel('|X(z)|');
title('Magnitude of Z-transform');

// Plot phase
subplot(2,1,2);
plot(theta, phase, 'r');
xlabel('Theta (radians)');
ylabel('Angle(X(z))');
title('Phase of Z-transform');

```

Graphic window number 0

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